

DAY 4 FIREWALL BASICS THEORY

Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

ANS:

Step 1: Press the Windows Key, type **Windows Defender Firewall with Advanced Security**, and open the application.

Step 2: Click on **Outbound Rules** in the left-hand panel, then select **New Rule...** from the right-hand actions menu.

Step 3: Select **Program** as the rule type, browse to locate the Spotify executable path (**Spotify.exe**), and click **Next**.

Step 4: Choose **Block the connection**, click **Next**, check all network profile boxes (Domain, Private, Public), and name the rule (e.g., **Block Spotify Outbound**) before clicking **Finish**.

Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

ANS:

Step 1: Open **Windows Defender Firewall with Advanced Security** from the Start menu.

Step 2: Click on **Inbound Rules** in the left sidebar, then select **New Rule...** from the actions pane.

Step 3: Select **Program**, click **Next**, browse to locate the Google Chrome executable (**chrome.exe**), and click **Next**.

Step 4: Choose **Allow the connection**, click **Next**, select your network profiles, and name the rule (e.g., **Allow Chrome HTTPS**) before clicking **Finish**.

Step 5: (*Optional Advanced Filter*) Open the newly created rule's **Properties**, go to the **Protocols and Ports** tab, set the protocol type to **TCP**, and specify **443** under the remote port options to strictly limit incoming secure traffic to HTTPS.

Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.

ANS:

Step 1: Open **Windows Defender Firewall with Advanced Security** from the Windows Start menu.

Step 2: Click on **Inbound Rules** (or **Outbound Rules**, depending on the direction you want to block) in the left panel.

Step 3: Click **New Rule...** in the right-hand actions pane to launch the wizard.

Step 4: Select **Custom** as the rule type and click **Next** (since blocking by specific IP requires custom rule settings).

Step 5: Choose **All programs** (or select a specific program if needed) and click Next, then leave the protocol and ports set to **Any** and click Next.

Step 6: Under the **Scope** settings for remote IP addresses, select **These IP addresses**, click **Add**, enter the IP address (123.45.67.89), click OK, and proceed.

Step 7: Select **Block the connection**, choose your applicable network profiles (Domain, Private, Public), and name the rule (e.g., **Block IP 123.45.67.89**) before clicking **Finish**.

Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

ANS:

Key Differences:

- **Scope:** Hardware firewalls protect an entire network perimeter, whereas software firewalls protect a single host device.
- **Performance:** Hardware firewalls use dedicated processors for high-throughput traffic, while software firewalls rely on the host machine's CPU/RAM.
- **Granularity:** Software firewalls can filter traffic down to specific local apps/processes, whereas traditional hardware firewalls focus on network/transport layers.

Use Cases:

- **Hardware Firewall:** Securing an enterprise office network edge.
- **Software Firewall:** Protecting a remote laptop on a public Wi-Fi network.